

# Exploring Emotional Intelligence Across Job Roles in Video Game Development Teams

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**Abstract**—The video game industry, an important sector within software development, relies on multidisciplinary teams to develop products that integrate a variety of media and technical elements. This complexity, coupled with high stress, highlights the importance of Emotional Intelligence (EI)—the ability to recognize, manage, and utilize emotions effectively. EI is crucial for leadership, team dynamics, and workplace well-being, yet variability in EI within teams can exacerbate workplace challenges. Despite its significance, research on EI remains limited. To address this gap, this study investigates EI across various job roles within the video game industry, examining both horizontal (role-specific) and vertical (hierarchical) differences. A survey using self-rated EI scales was administered to video game professionals. Statistical methods were used to analyze EI differences across roles and hierarchical levels. The analysis showed minimal differences in EI across horizontal roles. However, leaders exhibited higher scores in dimensions such as Managing One’s Own Emotions, indicating that EI is more pronounced in leadership roles. The findings suggest that, while the EI varies little by role, it is particularly relevant to leaders. Such a disparity between leaders and team members should not be taken lightly, as it could create an “emotional hierarchy,” potentially adding stress to leadership positions.

**Index Terms**—Emotional Intelligence, Video Game Development, Software Engineering, Socio-Technical Aspects

## I. INTRODUCTION

THE video game industry is one of the most prolific [1], [2], lucrative [2]–[4], and intricate sectors within software development [1], [5]. Its annual revenue has reached approximately USD 180 billion [2], [3], outpacing other entertainment industries [4]. Game development is a complex and multifaceted process [1], [5] that introduces distinct challenges [1], [5]–[7], with “crunch culture” representing one of the most critical and widely debated issues [8].

A central characteristic of video game development is its *multidisciplinary* nature [5], [9], [10]. Video games integrate diverse media—such as video, music, and narrative—necessitating collaboration among designers, developers, artists, and writers [5], [11]. Additionally, the multidisciplinary demands often attract geographically dispersed, culturally diverse individuals with varied personality traits [9], [12], increasing team heterogeneity. While heterogeneity can be beneficial, it also introduces socio-technical complexities [13]–[15] that can hinder collaboration [5], [11], [16].

An important factor with potential benefits in managing these complexities is **Emotional Intelligence (EI)** [17], [18]. EI—the ability to recognize, understand, manage, and use emotions effectively—has been increasingly recognized as essential for leadership [19], team dynamics [20], and workplace well-being [21]. Given the multidisciplinary and high-stress nature of video game development, substantial horizontal

(role-specific) and vertical (hierarchical) differences in EI are plausible and could impact productivity and team satisfaction. Furthermore, previous research indicates variations in personality traits across roles [12]—a variable often related to the ability to manage emotions—strengthening the need to explore emotional competencies within development teams.

Despite these observations, little research has characterized the EI of stakeholders throughout the game development lifecycle. Existing studies have focused primarily on end users (i.e., gamers) [22]–[26], leaving the development teams understudied. Investigating EI across roles presents a promising research opportunity, as emotional competencies may influence stress culture and socio-technical challenges within teams [20], [21], [27], [28]. Confirming such findings could serve as the “kiln of the first flame,” sparking further investigation and guiding research aimed at providing practitioners with strategies to address the complex and often problematic dynamics reported within video game development teams.

**This study aims to address this gap by characterizing Emotional Intelligence across different job roles in video game development, examining potential differences along horizontal (role-specific) and vertical (hierarchical) dimensions.** A survey was conducted to gather self-assessed EI scores and demographic information from active game development professionals. Statistical analyses were applied to determine whether significant differences in EI exist across role categories.

This work aims to contribute to the research field by highlighting EI—an often underrated factor in the video game industry—and by providing practitioners with actionable insights to address it more effectively, while offering researchers new directions to advance the state of the art.

## II. BACKGROUND AND RELATED WORK

This section describes the background and related work that is the foundation for the contribution.

### A. Emotional Intelligence

Emotional Intelligence (EI) is broadly defined as the capacity to recognize, understand, manage, and utilize emotions effectively in oneself and others [17], [18]. Originally conceptualized as a set of emotional and social skills, EI has been theorized to significantly impact various aspects of personal and professional life, from interpersonal communication to decision-making. Goleman [29] popularized the concept, suggesting that EI may be as critical as cognitive intelligence for success across diverse domains.

The Mayer and Salovey’s theory [17], [18] conceptualizes EI as a series of four interrelated abilities: perceiving, using, understanding, and managing emotions. *Perceiving emotions* involves accurately identifying emotions in oneself and others, whether through facial expressions, tone of voice, or other cues. *Using emotions* pertains to harnessing emotional states to facilitate cognitive processes, such as problem-solving and decision-making. *Understanding emotions* includes recognizing the relationships between emotional states and their evolution over time, as well as knowing the causes and consequences of different emotional experiences. Finally, *managing emotions* reflects the capacity to regulate both one’s own emotions and those of others, facilitating socially appropriate behavior and emotional expression.

Extensive research on EI demonstrates its role in enhancing leadership effectiveness, team performance, and employee well-being [30], [31]. Studies consistently show that individuals with high EI experience improved social relationships, job satisfaction, and resilience under stress [32]. For instance, Cherniss [33] found that EI training programs can significantly improve job performance and reduce turnover, underscoring EI’s practical relevance in professional settings. Such benefits are particularly critical in multidisciplinary, high-pressure environments like video game development teams, where emotional regulation and effective interpersonal communication are essential to sustain collaboration and productivity.

In the software development field—particularly, the software engineering (SE) and management ones—research on EI remains sparse. Kosti et al. [28] revealed connections between developers’ EI and their work preferences, indicating that individuals with higher EI prefer managing the entire development process independently and prioritizing tasks themselves rather than relying on managerial guidance. Additionally, Rezvani [34] demonstrated that higher EI negatively correlates with stress and promotes trust among developers. More recently, Madampe et al. [27] emphasized the central role of EI in managing software requirements changes, identifying EI as crucial in handling the emotional demands of such tasks. Similar emotional demands are expected to arise in video game development, given its multidisciplinary structure, creative pressures, and the tight deadlines often characteristic of the industry. Despite these findings, the research community continues to call for further investigation into EI dimensions within software development teams. In particular, there is a need to consider practitioners in the video game development industry, a highly challenging and lucrative sector within software development. While EI has been studied among gamers (i.e., end-users of video games), little attention has been paid to its influence on game developers themselves [22]–[24]. This gap in the literature motivates the present investigation into Emotional Intelligence across different job roles and hierarchical levels within video game development teams.

### B. Video Game Development and Job Roles

A video game is an interactive digital experience that blends art, narrative, music, and technology to captivate and entertain players [10]. For such a reason, video game development itself

is inherently multidisciplinary, requiring collaboration among designers, developers, artists, writers, and others to integrate diverse media and perspectives [5], [9], [10]. This process introduces unique challenges, such as the “crunch culture” that pervades the industry and has drawn attention within academia and the industry alike [8].

Moreover, the landscape of job roles within video game development reflects a sophisticated blend of occupational identity, specialized engineering practices, and shifting educational demands. Developers in this field constitute a distinctive occupational community, where identity formation and career progression are shaped by the industry’s project-based, boundaryless nature [35]. This professional identity is influenced by unique cultural production processes and a lack of standardized crediting, resulting in a complex structure of norms and competencies that shape the behavior and expectations of game developers [36].

In order to respond to such a complex and intriguing field, *Game Software Engineering (GSE)* has emerged as an independent discipline. The field of GSE continues to expand, accommodating industry-scale games through a refined approach to software development that acknowledges the specific demands and workflows inherent to game production [9], [16], [37]. Understanding the emotional competencies required by different job roles in Game Software Engineering contexts could therefore provide valuable insights for fostering better collaboration, resilience, and well-being in multidisciplinary development teams.

More on the human and collaborative aspects of (Game) Software Engineering, Sturdee et al. [12] examined the personality traits of game developers using the Five Factor Model, revealing that they differ significantly from other software professionals. The study, involving 123 developers, found that game developers tend to have lower levels of Openness, Conscientiousness, Extraversion, and Agreeableness, and higher Neuroticism, particularly among programmers. These personality traits may be shaped by the high-stress “crunch” culture typical of the industry. The authors suggest that understanding these differences can help improve team dynamics, recruitment, and stress management practices.

Addressing the complexity of job roles in video game development, Politowski et al. [6], [7] and Ullmann et al. [16] conducted analyses of video game postmortems, revealing numerous challenges within the socio-technical dynamics of development teams. Their findings emphasize the need for further research to support practitioners in overcoming these issues.

Overall, persistent challenges underscore the necessity of foundational research in the growing field of GSE [37]. Building on this foundational work, this study specifically investigates how Emotional Intelligence may vary across job roles and leadership positions in game development teams, aiming to contribute new insights to the human aspects of Game Software Engineering. Collaborative and socio-technical studies are critical for addressing the intricate human interactions within video game development teams [6], [7], [27], enhancing our understanding of this specialized and complex environment.

### III. OBJECTIVE AND HYPOTHESIS

To contribute addressing the identified gaps, the primary goal of this study is to investigate whether EI varies across different job roles within video game software development teams. Specifically, this study examines two dimensions of job roles: *horizontal*, referring to role-specific distinctions (e.g., developers, designers, and artists at the same level), and *vertical*, referring to roles at different levels within the organizational hierarchy (e.g., team members vs. managers). The *purpose* is to explore this relatively uncharted field, aiming to provide insights that support practitioners in managing resources effectively and making informed decisions, while enabling researchers to expand their understanding of video game development teams and to conduct further research focused on enhancing socio-technical well-being.

The hypothesis that inspired this work is the following:

*Emotional Intelligence varies significantly across both horizontal (role-specific) and vertical (hierarchical) job roles in video game development teams.*

This hypothesis is grounded in the notion that differing responsibilities, interpersonal demands, and decision-making pressures across roles may cultivate distinct EI competencies, shaping how individuals manage emotions and navigate collaborative challenges within their specific job functions and hierarchical positions [19]–[21]. Additionally, the multidisciplinary nature of video game development further supports the likelihood of individuals in different positions exhibiting distinct characteristics [5], [9]. Recent studies have also highlighted that personality traits—often linked to EI—vary across stakeholders within video game development teams, underscoring the need for further investigations [12]. However, this hypothesis remains debatable, particularly regarding the horizontal dimension. Given that video game development is a multidisciplinary field, it is possible that individuals tend to occupy multiple roles simultaneously [9], [38]; for example, roles such as designer and writer are frequently handled by the same individuals. In such cases, distinctions across horizontal roles may be less pronounced.

Based on the goal outlined above, two research questions (RQs) were formulated to guide the investigation, each targeting a distinct dimension of job roles. The first question seeks to understand differences in EI across same-level job-specific roles (e.g., developers, designers, and artists), while the second focuses on the hierarchical dimension (e.g., junior designer vs. lead designer). The first dimension is of interest as it captures the multidisciplinary nature of video game development, enabling an investigation that complements Sturdee et al.’s [12] work on personality traits. In contrast, the second dimension focuses on the established correlation between EI and leadership skills, exploring how EI may vary across hierarchical levels within teams.

**❓ RQ<sub>1</sub>** — *Is Emotional Intelligence significantly different across same-level job-specific roles (horizontal level) within video game development teams?*

**❓ RQ<sub>2</sub>** — *Is Emotional Intelligence significantly different across hierarchical job roles (vertical level) in video game development teams?*

The two RQs are formalized with the following hypotheses:

- *Null Hypotheses*: Emotional Intelligence is not significantly different across (**H1<sub>0</sub>**) same-level job-specific roles (horizontal level) and (**H2<sub>0</sub>**) hierarchical levels (vertical level) within video game development teams.
- *Alternative Hypotheses*: Emotional Intelligence is significantly different across (**H1<sub>A</sub>**) same-level job-specific roles (horizontal level) and (**H2<sub>A</sub>**) hierarchical levels (vertical level) within video game development teams.

### IV. RESEARCH DESIGN

In order to test the hypotheses, a survey study was conducted, following these steps:

- 1) First, the target population was identified and a sample strategy was defined to identify a reliable and representative set of participants;
- 2) Second, a questionnaire was developed, aimed at collecting demographic information from participants and EI values (used validated self-rated scales);
- 3) Third, the questionnaire was administered to the sample and data were collected;
- 4) Last, statistical tests were used for analysis.

#### A. Participant Selection and Criteria

The target population was composed of practitioners engaged in video game development. The *Prolific Academic* platform<sup>1</sup> was utilized to identify and recruit the sample, leveraging its purposive and cluster sampling capabilities to implement specific selection criteria. Guided by the methodology of Baltes and Ralph [39] and with the intent to comprehensively represent the population of video game practitioners, only a single primary criterion was applied: *current or recent (within the past year) involvement in a video game development team*.<sup>2</sup>

Additional filters were applied to enhance participant reliability and comprehension. Participants were required to be fluent in English (the survey language), maintain a 100% approval rate for previous studies, and have completed at least 25 surveys on Prolific. The approval rate metric served as an indicator of participant reliability, reflecting consistency and quality in prior responses. Moreover, responses that failed attention checks (e.g., “select the third option” in a multiple answer matrix) or were completed in under half the estimated time were excluded from the dataset. To further enhance sample quality, I applied a full-time employment filter in Prolific. Additionally, I designed a set of domain-specific screening questions, inspired by Danilova et al. [40], to assess participants’ familiarity with the video game industry. Participants were asked to correctly answer three questions regarding gaming platforms and engines before proceeding with the survey (e.g., “Which of the following is NOT a

<sup>1</sup>Prolific (www.prolific.com) [October 2024]

<sup>2</sup>Participants received an informed consent form before agreeing to participate in the study.

gaming platform? (1) Origin (EA), (2) Playstation Network, (3) GigaPlay, and (4) Epic Games Store”). Only those passing all screening questions were retained.

### B. Participants Sample

Prolific provided a potential pool of 224 509 participants. After applying the specified filters, this pool was reduced to 3114 participants. Following a preliminary exclusion for attention check failures, 207 responses were retained. Finally, participants with excessively fast completion times were removed, resulting in a final dataset of 186 accepted responses.

The participants demonstrates varied levels of experience. Most participants report between 1-3 years in their current roles (39.5%) and in the video game industry overall (44.3%). Those with 4-6 years of experience comprise 24.3% in their roles and 21.6% in the industry. A smaller subset has significant experience of over nine years in both categories, representing 11.9% in roles and 13.5% in the industry.<sup>3</sup>

The organizational size distribution indicates that many participants work in smaller organizations, with 36.8% in organizations of 1-25 members and 31.9% in organizations with 25-100 members. Larger organizations (100-500 and over 500 members) represent 20.5% and 11.4% of the sample, respectively.

Demographically, the sample skews male, with 68.1% identifying as male and 30.8% as female. The age distribution is broad, with a mode of 26 years, situating most participants in the young adult range. Geographically, the sample shows a substantial representation from Europe, with 102 participants (55.1% of the sample). South Africa follows with 32 participants (17.3%). Canada (17 participants, 9.2%) and the United States (10 participants, 5.4%) are also represented, with smaller groups from countries such as Chile and Mexico.

### C. Questionnaire Design and Measurement Instruments

The questionnaire was designed following best practices in software development research [41], [42]; both surveys were anonymous and included introductory descriptions, a final question inviting feedback, and three attention-check items to enhance response reliability.

An iterative pilot testing strategy was used to ensure quality and clarity. First, two pilot tests were conducted with 5 researchers, with feedback incorporated to improve survey comprehension and estimate completion time. Subsequently, pilot tests were conducted directly on Prolific with a small sample of qualified participants. No further modifications were needed, and pilot responses were included in the final dataset.

The measurement and data collection focused on three main variables: participants’ EI scores, their job roles, and demographic information, all of which were instrumental in characterizing the sample.

<sup>3</sup>I intentionally retained participants with less experience to reflect the natural heterogeneity of development teams, which often include junior developers, interns, and early-career professionals. Moreover, given that EI develops over time and experience, including younger professionals provided a broader and more realistic depiction of industry demographics.

For EI measurement, the self-reported EI scale (SREIS) developed by Brackett et al. [43] (and based on well-known scales in the literature [44], [45]) was selected due to its good psychometric reliability and the condition of this investigation (recruitment methods and resource constraints). The scale operationalizes EI through the five dimensions of EI theory: Perceiving Emotion (P), Use of Emotion (U), Understanding Emotion (R), Managing One’s Own Emotions ( $M_1$ ), and Managing Others’ Emotions ( $M_2$ ). This 19-item scale employs a 5-point Likert format from “Very Inaccurate” to “Very Accurate.” Example items include: “By looking at people’s facial expressions, I recognize the emotions they are experiencing” (P), “When making decisions, I listen to my feelings to see if the decision feels right” (U), “I could easily write a lot of synonyms for emotion words like happiness or sadness” (R), “I can handle stressful situations without getting too nervous” ( $M_1$ ), and “I know the strategies to make or improve other people’s moods” ( $M_2$ ).

Participants also reported their job roles. Recognizing that video game development could involve individuals undertaking multiple roles at the same time [9], [38], participants were permitted to select multiple job roles. Building on prior research, the categorization used by Sturdee et al. [12] was adopted, with minor roles added from both literature [9] and pilot study findings. The roles included were:

- Production and Management (Production): Oversee timelines, coordinate team efforts, and ensure project milestones align with objectives.
- Design and Narrative (Design): Develop game mechanics, structure narratives, and establish the player experience.
- Programming and Engineering (Programming): Write code, develop systems, and implement game features, focusing on performance and functionality.
- Data and Quality Assurance (QA): Conduct testing to identify issues, analyze data for insights, and ensure the game meets quality standards.
- Artists and Animators (Art): Create visual assets, animations, and bring characters and environments to life.
- Audio and Music (Audio): Develop soundtracks, sound effects, and voiceovers.

For the second research question, participants were also asked to indicate their hierarchical level within each role. For example, those identifying as “Designers” were asked to specify whether they were junior designers or lead designers.

The questionnaire included various demographic data and confounding variables, modeled after Sturdee et al. [12]. These encompassed age, gender, years of experience in their current role and in the video game industry, company size, and primary development platform.

### D. Data Analysis

The analysis commenced with a preliminary examination of the dataset to ensure data quality. The dataset was imported into JASP [46], an open-source platform for statistical analysis, and preliminary descriptive statistics were conducted to assess normality. Normality was evaluated through kurtosis, skewness, and the Shapiro–Wilk test. The results indicated that

the data did not follow a normal distribution in most cases, thus informing the decision to adopt non-parametric tests for subsequent analyses.

Using JASP, a series of reliability tests were conducted to assess the validity of the questionnaire-based measurement. Specifically, Cronbach's  $\alpha$ , Exploratory Factor Analysis (EFA), and Confirmatory Factor Analysis (CFA) were employed. These validated techniques provided a comprehensive evaluation of the measurement process and confirmed its overall adequacy [47].

For the first RQ, a considerable number of participants were expected to report more than one job role in the survey. Given the objective of analyzing role-specific influences on EI, replicating entries for participants with multiple roles was a reasonable approach; this method preserves the multidimensional nature of participants' professional identities, allowing the model to capture variations in EI across different roles without forcing a reduction to a single, primary role. Kruskal-Wallis test was used for the analysis. This non-parametric test was selected as it does not require assumptions of normality and is robust to unequal group sizes, making it suitable for the structure of the dataset.<sup>4</sup>

For the second research question, since the vertical dimension of job roles was represented as a two categories variable (leader and non leader), a Mann-Whitney test was used. Specifically, I divided the roles into two groups: the first group includes all leadership and management roles, such as Design Lead and Artist Lead; the second group comprises all non-leader roles, which are the remaining positions. I then conducted the analysis using these two groups as treatment variables. This non-parametric alternative to the t-test was appropriate given the ordinal nature of the data and the lack of normality in the distribution of the measured variables.

Additionally, although this study primarily aimed to investigate significant differences rather than establish causation (which would require a more systematic approach and a broader research trajectory), several confounding variables were considered, including years of experience, gender, and age [49], [50]. To analyze the potential influence of these confounders, it was planned to implement a linear regression model, incorporating the dependent variable alongside all confounders in cases where the statistical tests highlighted previously revealed significant differences. In case of assumption violations, bootstrapping version of the model (5000 iterations) was planned to be used.<sup>5</sup>

## V. RESULTS

In this section, the results of the analysis are reported.

<sup>4</sup>A mixed-effects model [48] was initially chosen, with Participant ID as a random effect. However, warnings about numerical issues in the Maximum-Likelihood Estimate, specifically large gradients, indicated difficulties due to the limited sample size relative to model complexity.

<sup>5</sup>The choice of the model was guided by (1) the continuous dependent variable, (2) the ordinal nature of most of the dependent/confounding variable, and (3) the violation of assumptions for standard models. For space limitations, the detailed results of the confounding analysis are available in the online appendix (doi:10.6084/m9.figshare.27636741).

TABLE I  
ITEMS CRONBACH'S  $\alpha$ .

| Factor                                        | Item ID | Cronbach's $\alpha$ |
|-----------------------------------------------|---------|---------------------|
| Perceiving Emotion (P)                        | P1      | 0.783               |
|                                               | P2      | 0.793               |
|                                               | P3      | 0.783               |
|                                               | P4_r    | 0.795               |
| Using Emotions (U)                            | U1_r    | 0.816               |
|                                               | U2      | 0.796               |
|                                               | U3      | 0.813               |
| Understanding Emotions (R)                    | R1      | 0.782               |
|                                               | R2      | 0.778               |
|                                               | R3      | 0.779               |
|                                               | R4      | 0.790               |
| Managing One's Own Emotions (M <sub>1</sub> ) | M1_1_r  | 0.799               |
|                                               | M1_2    | 0.789               |
|                                               | M1_3    | 0.792               |
|                                               | M1_4    | 0.789               |
| Managing Others' Emotions (M <sub>2</sub> )   | M2_1    | 0.779               |
|                                               | M2_2    | 0.783               |
|                                               | M2_3_r  | 0.798               |
|                                               | M2_4    | 0.786               |
| Overall Cronbach's $\alpha$                   |         | 0.800               |

### A. Measurement and Scales Validation

The first step in validating the measurement involved calculating Cronbach's  $\alpha$  (reported in Table I). All values exceeded 0.7, indicating good reliability of the items.

The next step involved conducting EFA. Prior to factor analysis, Bartlett's Test of Sphericity and the Kaiser-Meyer-Olkin (KMO) Test for sampling adequacy were applied to assess data factorability. Bartlett's Test was significant ( $p < 0.001$ ), indicating sufficient interrelationships between the items. KMO values were satisfactory for most items, with an overall KMO score of 0.791, confirming that the data was suitable for factor analysis. EFA was conducted using the eigenvalue rule and the *promax* option to determine the number of factors. The analysis identified six factors, one more than the five factors proposed in the EI theory. Despite this, items loaded distinctly onto separate factors, with factor loadings close to 0.3 for most items, indicating adequate item-factor associations. One item from the U scale (U3) reported a loading below 0.3 ( $|0.278|$ ). However, as highlighted by Brackett et al. [43], assessing the use of emotions is complex, requiring a nuanced combination of multiple items; the final decision of retain the item was moved to the CFA step.

For CFA, the initial model showed acceptable fit indices: a Comparative Fit Index (CFI) of 0.903, a Tucker-Lewis Index (TLI) of 0.883, and a Root Mean Square Error of Approximation (RMSEA) of 0.063. To further improve the model and retain at least three items per factor, two items (P4\_r and M1\_1\_r) were removed. This adjustment led to improved fit measures, resulting in a final CFI of 0.937, TLI of 0.921, and RMSEA of 0.055.

### B. RQ1—Job Roles and Emotional Intelligence

This section presents the findings from the analysis of EI dimensions across different (horizontal) job roles within video game development teams.

TABLE II  
DESCRIPTIVE STATISTICS FOR THE FIRST RQ

| Factor         | Job               | #  | Mean  | St. Dev. |
|----------------|-------------------|----|-------|----------|
| P              | Audio             | 11 | 3.939 | 0.554    |
|                | Production        | 32 | 3.865 | 0.528    |
|                | Design            | 44 | 3.962 | 0.572    |
|                | Art               | 23 | 3.855 | 0.758    |
|                | Quality Assurance | 29 | 3.920 | 0.621    |
|                | Programming       | 72 | 3.824 | 0.773    |
| U              | Audio             | 11 | 3.212 | 0.402    |
|                | Production        | 32 | 3.635 | 0.613    |
|                | Design            | 44 | 3.492 | 0.625    |
|                | Art               | 23 | 3.391 | 0.398    |
|                | Quality Assurance | 29 | 3.644 | 0.519    |
|                | Programming       | 72 | 3.500 | 0.525    |
| R              | Audio             | 11 | 3.273 | 0.737    |
|                | Production        | 32 | 3.875 | 0.699    |
|                | Design            | 44 | 3.994 | 0.696    |
|                | Art               | 23 | 3.163 | 0.764    |
|                | Quality Assurance | 29 | 3.741 | 0.944    |
|                | Programming       | 72 | 3.608 | 0.849    |
| M <sub>1</sub> | Audio             | 11 | 3.394 | 1.042    |
|                | Production        | 32 | 3.896 | 0.726    |
|                | Design            | 44 | 3.788 | 0.705    |
|                | Art               | 23 | 3.406 | 0.724    |
|                | Quality Assurance | 29 | 3.839 | 0.839    |
|                | Programming       | 72 | 3.644 | 0.836    |
| M <sub>2</sub> | Audio             | 11 | 3.523 | 0.684    |
|                | Production        | 32 | 3.938 | 0.629    |
|                | Design            | 44 | 3.886 | 0.559    |
|                | Art               | 23 | 3.663 | 0.697    |
|                | Quality Assurance | 29 | 3.655 | 0.800    |
|                | Programming       | 72 | 3.594 | 0.893    |

TABLE III  
KRUSKAL-WALLIS TEST FOR THE FIRST RQ

| Factor                                      | Statistic | $p$     | Rank $\epsilon^2$ |
|---------------------------------------------|-----------|---------|-------------------|
| Perceiving Emotion (P)                      | 0.498     | 0.992   | 0.002             |
| Using Emotions (U)                          | 7.963     | 0.158   | 0.038             |
| Understanding Emotions (R)                  | 21.194    | < 0.001 | 0.101             |
| Managing Own Emotions ((M <sub>1</sub> ))   | 9.768     | 0.082   | 0.047             |
| Managing Others' Emotions (M <sub>2</sub> ) | 5.467     | 0.362   | 0.026             |

An initial analysis showed that 123 participants selected a single role, 44 selected two roles, and 19 selected more than two roles. To maintain consistency and due to the limited number of participants with more than two roles, only individuals with up to two job roles were included in the analysis for the first RQ.<sup>6</sup> Table II provides summary statistics for each job role, which serve as a foundation for interpreting the results. The Kruskal-Wallis test (reported in Table III), was employed to identify significant differences across job roles, with Dunn's post hoc tests conducted for dimensions showing statistically significant results. As observed, only one variable, R, showed statistically significant differences. **Therefore, hypothesis H1<sub>A</sub> is not supported.**

Regarding *Understanding Emotions (R)*, the test revealed statistically significant differences across job roles ( $p <$

<sup>6</sup>To ensure transparency and robustness, two parallel analysis were conducted: one with all participants and one with only those who identified with a single job position. Both analyses yielded the same results of the analysis reported in the section.

TABLE IV  
INDEPENDENT SAMPLES T-TEST FOR THE SECOND RQ

| Factor         | $p$   | Hodges-Lehmann Est.    | Rank-Biserial Corr. |
|----------------|-------|------------------------|---------------------|
| P              | 0.122 | $6.803 \times 10^{-6}$ | 0.146               |
| U              | 0.037 | 0.333                  | 0.196               |
| R              | 0.036 | 0.250                  | 0.199               |
| M <sub>1</sub> | 0.012 | 0.333                  | 0.237               |
| M <sub>2</sub> | 0.052 | 0.250                  | 0.185               |

0.001), with post hoc comparisons indicating specific contrasts: Production vs. Art ( $p = 0.002$ ) and Design vs. Art ( $p < 0.001$ ) remained significant even after Bonferroni and Holm corrections, highlighting notable differences in emotional understanding skills between these roles. Design and Production roles generally exhibited higher R scores, while Art tended toward lower scores. These findings imply that the ability to understand emotions is significantly higher among individuals in Design and Production roles. A subsequent analysis conducted with the confounding variables revealed that none of them (i.e., age, gender, and years of experience) had a significant impact on the sample compared to the job role, which remained significant even in this context. This allows for greater confidence in the results.

Focusing on *Managing One's Own Emotions (M<sub>1</sub>)*, the Kruskal-Wallis test approached significance ( $p = 0.082$ ), suggesting a trend toward differences across job roles. The highest mean score was observed for the Production role (3.896), while the lowest was for Art (3.406). This trend, though not statistically robust, may indicate variations in self-management of emotions based on role responsibilities.

### III RQ<sub>1</sub> — Summary of the Results

Statistically significant job role differences were observed only for the capacity of **understanding emotions**, indicating higher emotional regulation skills in roles such as Design and Production. The marginal trends in managing one's own emotions suggest possible differences that may emerge with a larger sample size.

### C. RQ<sub>2</sub>—Leading Job Roles and Emotional Intelligence

The analysis, summarized in Table IV, reports the results of independent samples T-tests (Mann-Whitney U tests) comparing the emotional intelligence dimensions between leaders (50 occurrences) and non-leaders (136 occurrences) within video game development teams. Significant differences emerged in three of the five dimensions: U, R, and M<sub>1</sub>. The M<sub>2</sub> dimension approached significance, whereas P did not exhibit a significant difference. **These findings provide substantial support for H2<sub>A</sub>.** Moreover, in this case as well, the use of a linear regression model, combined with bootstrapping to address the violation of assumptions for the standard version of the model, revealed that for none of the dependent variables did the confounding variables have a greater (or even significant) impact compared to the independent variable. Figure 1 presents violin plots illustrating the data distribution; each variable is analyzed in detail below.

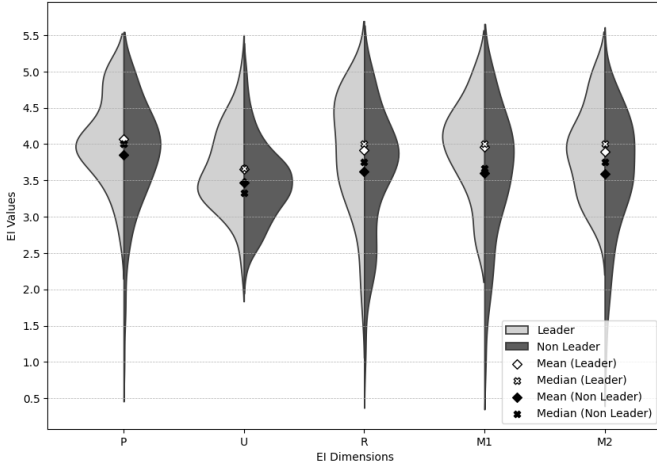


Fig. 1. EI values distributed according to dimensions and vertical roles.

1) *Perceiving Emotion (P)*: The ability to recognize and interpret others' emotions, does not show a statistically significant difference between leaders and non-leaders ( $p = 0.122$ ). The Hodges-Lehmann Estimate is very small ( $6.803 \times 10^{-6}$ ), and the Rank-Biserial Correlation of 0.146 indicates a weak effect size. The violin plot reveals similar distributions. Moreover, it shows a minimal difference in mean scores (leaders: 4.07, non-leaders: 3.85). This suggests that P may be equally developed across both groups.

2) *Using Emotion (U)*: The U dimension, which assesses individuals' reliance on emotions in decision-making, shows a statistically significant difference between leaders and non-leaders ( $p = 0.037$ ). The Hodges-Lehmann Estimate of 0.333 suggests a moderate difference in central tendency, and the Rank-Biserial Correlation of 0.196 indicates a small-to-moderate effect size. The violin plot reveals a slightly higher median for leaders with less variability and illustrates a clear difference in mean scores (leaders: 3.66, non-leaders: 3.47). These findings imply that leaders may incorporate emotional insights more effectively into their decision-making processes.

3) *Understanding Emotion (R)*: In the R dimension, which involves comprehending and articulating emotions, leaders scored significantly higher than non-leaders ( $p = 0.036$ ). The Hodges-Lehmann Estimate of 0.250 reflects a moderate difference, and the Rank-Biserial Correlation of 0.199 further supports a small-to-moderate effect size. The violin plot indicates a higher central tendency for leaders, with less variability compared to non-leaders (leaders: 3.92, non-leaders: 3.62). This pattern suggests that leaders in video game development teams may possess a more advanced ability to understand emotional nuances.

4) *Managing One's Own Emotions (M1)*: Regarding the ability to regulate one's own emotional responses (M1), it is significantly different between the two groups ( $p = 0.012$ ). The Hodges-Lehmann Estimate of 0.333 reflects a meaningful difference, and the Rank-Biserial Correlation of 0.237 suggests a moderate effect size. The violin plot shows a higher median and narrower spread for leaders, indicating more consistent scores. It also indicate a difference in means (leaders: 3.96,

non-leaders: 3.6). These findings imply that leaders are generally more skilled at managing their own emotions.

5) *Managing Others' Emotions (M2)*: Regarding the other side of managing emotions, i.e., managing others' emotions (M2), it approaches statistical significance ( $p = 0.052$ ), with a Hodges-Lehmann Estimate of 0.250 and a Rank-Biserial Correlation of 0.185. The violin plot shows a marginally higher median for leaders with considerable overlap between groups; it also reflects a slight difference in means (leaders: 3.9, non-leaders: 3.59). Although the difference is not statistically significant, the trend suggests that leaders might be marginally better at managing others' emotions.

## RQ2 — Summary of the Results

Leaders in video game development teams score higher on **using, understanding, and managing their own emotions**, highlighting these skills as potential factors in effective leadership within these teams. Managing others' emotions shows a near-significant trend, suggesting a potential area for further exploration.

## VI. DISCUSSION AND IMPLICATIONS

This section provides a contextual analysis of the findings and highlights potential implications for both practitioners (indicated using the 🧑 symbol) and researchers (indicated using the 🔍 symbol).

### A. On the Differences Between Horizontal Job Roles

This study revealed limited variation in EI across horizontal roles, with only the ability to understand emotions showing significant differences. Unlike personality traits, which have been shown to vary by role [12], EI appears less influenced by role-specific factors in game development. This suggests that EI may be shaped more by individual training and experience than by inherent traits, highlighting its potential to be cultivated across diverse roles.

🔍 **Future research should explore the relationship between EI and other psychometric traits.** Future studies could delve deeper into the relationship between EI and personality to better understand how these constructs interact within multidisciplinary teams. Additionally, cultural intelligence [51] has been shown to correlate with both personality traits and EI [52]; investigating this relationship further could provide valuable insights, particularly given the inherently distributed nature of the game industry [53].

Notably, Design and Production roles demonstrated higher scores in understanding emotions compared to Art roles. This disparity likely reflects the distinct responsibilities of Design and Production, which require greater emotional comprehension to manage workflows and facilitate collaboration in multidisciplinary teams [27], [28]. In contrast, Art roles, often emphasizing individual creativity over team-oriented engagement, may place less emphasis on emotional understanding.

Although trends suggested slightly higher self-regulation scores for Production roles, differences in other EI dimensions, such as perceiving and managing emotions, were not significant. This uniformity across roles implies that emotional competencies may be broadly distributed within game development, reflecting their general applicability rather than role-specific demands. Further research is needed to explore the interplay between EI and personality traits and to investigate the unique challenges faced by creative roles.

### B. On the Differences Between Vertical Job Roles

The analysis highlights significant disparities in EI dimensions between leaders and non-leaders, reflecting the heightened interpersonal and emotional demands of leadership in game development. These findings align with the EI theory proposed by Salovey and Mayer [17] and the Transformational Leadership theory proposed by Bass and Avolio [54], which emphasize the pivotal role of EI and personal skills in leadership. Furthermore, the differences observed between leadership and non-leadership roles in terms of EI may also be interpreted through the lens of organizational context. Prior research has shown that indie studios often operate with fluid, overlapping, and informally negotiated roles, where practitioners engage in entrepreneurial and relational labor [55]. In contrast, AAA studios tend to maintain hierarchical structures with more narrowly defined responsibilities. These distinctions imply that EI may not only vary across formal roles, but also according to how those roles are enacted in different production settings. For instance, in indie teams, leadership responsibilities may emerge from collaboration and relational dynamics rather than formal titles, potentially requiring different emotional competencies.

Leaders' elevated scores in *Understanding Emotions (R)* reflect their ability to navigate complex interpersonal dynamics and foster collaboration within multidisciplinary teams. These skills are particularly critical in game development, where creative and technical contributions must align under tight deadlines [27], [28]. Similarly, the significant difference in *Using Emotions (U)* highlights leaders' ability to incorporate emotional insights into decision-making processes. This skill fosters more adaptive and holistic leadership, enabling leaders to balance technical objectives with the emotional needs of their teams [28].

Leaders' stronger abilities in *Managing One's Own Emotions (M<sub>1</sub>)* indicate their capacity to maintain composure and adaptability under pressure, a key asset during demanding periods like "crunch time" [8]. While beneficial for stabilizing team dynamics, this concentration of self-regulation skills within leadership may leave non-leaders ill-equipped to manage their own stress effectively. As noted by Madampe et al. [27], the absence of robust emotional regulation across all team levels could lead to imbalanced team dynamics and reduced productivity.

While *Managing Others' Emotions (M<sub>2</sub>)* approached significance, leaders' higher scores suggest that their roles inherently demand navigating team morale and resolving conflicts to sustain productivity. These skills are vital in fostering a

cohesive and supportive team environment, particularly during high-stress periods.

In contrast, no significant differences were observed in *Perceiving Emotions (P)*, suggesting that the ability to recognize and interpret emotions is equally distributed across hierarchical levels. This uniformity may reflect the universal importance of emotional perception in the multidisciplinary and collaborative nature of game development [9].

These findings highlight the dual-edged nature of concentrated EI competencies in leadership. Although they empower leaders to navigate complex team dynamics effectively, they also risk creating dependency structures that may strain leaders and limit team-wide emotional resilience. It is possible that this disparity can inadvertently establish an **"emotional hierarchy,"** where non-leaders may feel less equipped to handle emotional challenges independently. Such dependency risks overburdening leaders, potentially diminishing their well-being and increasing stress levels during critical project phases.

🔑 **Practitioners should closely monitor the emotional state of team leaders.** Although high EI is generally desirable, it may also increase vulnerability to stress under certain conditions. Implementing dedicated monitoring tools could help organizations detect early signs of overload and take preventive action, for instance by providing EI training across the team to reduce reliance on leaders.

💡 **Future research should explore EI's Impact on team metrics.** Given the observed differences between leaders and non-leaders, researchers could investigate how leaders' emotional management competencies affect team-level outcomes such as resilience, collaboration, or productivity. This line of inquiry could be especially relevant in high-stress domains such as game development.

### C. Limitations

As with any research, the findings presented in this article should be interpreted with certain limitations in mind.

Firstly, the results cannot be considered fully representative of the entire video game developer community. Although Prolific allows for reaching geographically distributed individuals with characteristics suited to the population under study, and despite a rigorous recruitment process, generalizability of the findings cannot be assured without reservation. Additionally, the demographic data showed a predominance of respondents from European countries, with underrepresentation from the Americas and Canada. Furthermore, despite implementing rigorous quality assurance procedures—including full-time employment filters, approval rate thresholds, attention checks, and domain-specific screening questions—I acknowledge the challenge in fully verifying the professional status of participants. Consequently, although this process effectively minimized such events, there is a possibility that some participants may have acted unethically.



💡 **Replications of the study.** Replications of this study are encouraged to either reinforce the generalizability of the findings or challenge them through alternative sampling strategies and contexts.

Related to the previous point, recruiting professional participants in software and game development studies is inherently difficult. As noted by Madampe et al. [56], even well-established empirical studies often rely on self-reporting and platform-based filters that cannot fully guarantee participant authenticity. In the context of the video game industry, Whitson et al. [55] highlight how non-standard roles, informal employment conditions, and entrepreneurial structures—particularly in indie game production—complicate the establishment of rigid professional boundaries. As such, I acknowledge that this study, despite using rigorous screening procedures, cannot entirely exclude the presence of borderline or atypical cases. Nonetheless, as mentioned above, the multi-layered filtering approach, including domain-specific screening questions and attention checks, mitigates these risks.

There is a limitation concerning the measurement scale, which may not have captured all facets of EI [43]. The chosen scale was previously validated and based on well-established instruments in the literature [44], [45], representing an ideal compromise for the requirements of this study. Furthermore, the rigorous process undertaken to ensure participant reliability and data accuracy supports the validity of the measurements. Nonetheless, measuring latent variables is inherently challenging, particularly in the case of EI [57], [58].

💡 **Future replications using different EI scales are encouraged.** The complexity of measuring certain EI dimensions suggests that future research could benefit from diverse EI scales or mixed-method approaches [57], [58]. Using varied measurement tools would help confirm or refine findings and provide a more comprehensive view of EI's impact on job roles.

To enhance the robustness and reliability of the results, confounding variables were considered both during participant selection and in the analysis phase. Naturally, it is not possible to account for all potential variables, as an overly exhaustive statistical analysis in this regard is prone to Type I errors [59]. Nevertheless, preliminary confounding analysis indicated that results are robust concerning age, gender, and years of professional experience—variables previously demonstrated to be associated with EI [49], [50]. Further investigation into additional variables is left to future studies, already underway. Until then, as with any statistical study, results should be interpreted conservatively and strictly within the context outlined in Section IV.

#### D. Future Research Directions

In addition to the previously discussed strategies for addressing this study's limitations, several additional directions for future research have been identified.

The software engineering research community has recently extended its focus on EI, employing mixed-methods approaches [27]. In particular, Madampe et al. [27] identified

EI as essential for managing changes in software requirements—a finding with considerable relevance in video game development as well. Conducting a similar investigation in this domain could yield valuable insights by adding a qualitative dimension, thereby enriching the current understanding. As an ulterior point, the socio-technical complexity inherent in software development [60], [61] is arguably heightened in video game development due to its pronounced multidisciplinary nature. Applying methodologies such as Hoda's Socio-Technical Grounded Theory [60] may be especially effective in this context, potentially uncovering new insights with greater clarity and depth.

💡 **Our quantitative findings could be corroborated with qualitative research.** Conducting an investigation similar to the one of Madampe et al. [27]—leveraging socio-technical grounded theory [60]—in the video game domain could provide valuable insights by adding a qualitative dimension, thus enriching current understanding.

The findings related to the second research question are consistent with established theoretical frameworks on EI [18] and leadership [54]. This alignment suggests the opportunity to adopt these theories and their latent constructs in more advanced quantitative investigations—such as those employing Partial Least Squares Structural Equation Modelling (PLS-SEM) [47]—to gain deeper insights into the dynamics of video game development.

💡 **Our future research agenda aims to further explore the role of EI through a theory-driven statistical lens.** Several theoretical models may help explain the observed differences in EI among video game practitioners by linking them to latent variables such as project success, leadership effectiveness, or well-being. In this context, PLS-SEM offers a robust framework for testing such hypotheses empirically.

Although this study examined EI across various job roles, it did not explicitly account for differences between small indie studios and larger AAA companies. However, this distinction may have significant implications for both the interpretation of job roles and the contextual demands placed on emotional skills. Prior research has highlighted that indie studios often adopt fluid and overlapping roles, with developers engaging in relational labor and entrepreneurial tasks beyond their formal titles [55]. In contrast, AAA companies typically adhere to more hierarchical and specialized structures.

💡 **Future work should consider stratifying analyses by studio size or production model.** The organizational divergences among video game teams may lead to different expressions and expectations of EI across contexts that deserve attention.

## VII. CONCLUSION

This study's findings reveal that, while EI does not significantly vary across job roles, certain EI dimensions—such

as Understanding Emotions and Managing One's Own Emotions—are more pronounced among leaders than non-leaders. This disparity suggests that leaders may bear a heavier emotional burden within teams, potentially contributing to the high stress levels often reported in the video game industry. The reliance on leaders to manage emotional dynamics could create an “*emotional hierarchy*,” where leaders are expected to handle interpersonal challenges, thereby increasing their risk of burnout. Nonetheless, this distribution of higher EI levels among leaders may be advantageous, as it equips those in leadership positions to navigate complex team dynamics and foster a supportive, cohesive work environment.

This paper advances the socio-technical research field in Game Software Engineering by:

- 1) offering a statistical overview of differences in EI among practitioners in the video game industry;
- 2) proposing a set of practical and research-oriented implications to support both practitioners and researchers;
- 3) making all materials available in an online appendix<sup>7</sup> to enhance replicability and reliability, as well as to encourage further research in this area.

For practitioners, the results suggest the potential benefits of EI training across all team levels, fostering a more emotionally resilient workforce. For researchers, these findings open avenues for further investigation into EI's impact on team dynamics and performance in creative and technical settings.

#### DATA AVAILABILITY

All data are available in an online appendix (doi:10.6084/m9.figshare.27636741). For ease of consultation, it is recommended to use the HTML files or the JASP tool for a more in-depth analysis.

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